

***Generation Interconnection
Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD2-079***

Capital Avenue 34.5 kV

November 2019

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: (1) Direct Connections, which are new facilities and/or facilities upgrades needed to connect the generator to the PJM network, and (2) Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement.

The System Impact Study estimates do not include the cost or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Indiana Michigan Power Company, proposes to install PJM Project #AD2-079, a 20.0MW (12.0 MW Capacity) Solar facility in St. Joseph county, IN (See figure 2). The point of interconnection will be a direct connection to AEP's Capital Avenue 34.5 kV substation (see Figure 1)

The requested in service date is November 1, 2020.

Attachment Facilities

Point of Interconnection (Capital Avenue 34.5kV)

To accommodate the interconnection on the Capital Avenue 34.5 kV station, the substation will have to be expanded requiring the installation of one 34.5 kV circuit breaker (see Figure 1). Installation of associated protection and control equipment, 34.5 kV line risers, SCADA, and 34.5 kV revenue metering will also be required.

Description	Estimated Cost
Install one 34.5kV Circuit Breaker at the Capital Avenue 34.5kV substation. Associated protection and control equipment, 34.5 kV line risers, SCADA, and 34.5 kV revenue metering will also be required.	\$500,000
34.5 kV Revenue Metering	\$200,000
Total	\$700,000

Table 1

Direct Connection at the Capital Avenue 34.5kV Substation work and cost:

None

Non-Direct Connection Cost Estimate

The total preliminary cost estimate for Non-Direct Connection work is given in the following table below:

For AEP building Non-Direct Connection cost estimates:

Description	Estimated Cost
Review and update Protection and Control settings on adjacent facilities.	\$50,000
Total	\$50,000

Table 2

Note: The 34.5kV facilities at Capital Avenue substation are being planned for future upgrades to 69kV. Many of the existing station facilities presently energized at 34.5kV are already designed for 69kV operation. Therefore, all the equipment and facilities to be added at Capital Avenue for the AD2-079 interconnection request must also be suitable for eventual 69kV operation.

Interconnection Customer Requirements

It is understood that Indiana Michigan Power Company is responsible for all costs associated with this interconnection. The cost of Indiana Michigan Power Company generating plant and the costs for the line connecting the generating plant to Indiana Michigan Power Company switching station are not included in this report; these are assumed to be Indiana Michigan Power Company's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

Network Impacts

The Queue Project AD2-079 was evaluated as a 20.0 MW (Capacity 10.0 MW) injection at the Capital Avenue 34.5kV substation in the AEP area. Project AD2-079 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-079 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Base Case Used

Summer Peak Analysis – 2021 Case

Contingency Descriptions

The following contingencies resulted in overloads:

None

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

No problems identified.

Affected System Analysis & Mitigation

LGEE Impacts:

None

MISO Impacts:

None

Duke, Progress & TVA Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Schedule

It is anticipated that the time between receipt of executed agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would be between 24 to 36 months after signing an interconnection agreement.

Note: The time provided between anticipated normal completion of System Impact, Facilities Studies, subsequent execution of ISA and ICSA documents, and the proposed Backfeed Date is shorter than usual and may be difficult to achieve.

Conclusion

Based upon the results of this Impact Study, the construction of the 20.0 MW (12.0 MW Capacity) solar generating facility of Indiana Michigan Power Company (PJM Project #AD2-079) will require additional interconnection charges. This plan of service will interconnect the proposed solar generating facility in a manner that will provide operational reliability and flexibility to both the AEP system and the Indiana Michigan Power Company.

Cost Breakdown for Point of Interconnection (Capital Avenue 34.5kV)		
Attachment Cost	Install one 34.5kV Circuit Breaker at the Capital Avenue 34.5kV substation. Associated protection and control equipment, 34.5 kV line risers, SCADA, and 34.5 kV revenue metering will also be required.	\$500,000
	34.5 kV Revenue Metering	\$200,000
Non-Direct Connection Cost Estimate	Review and update Protection and Control settings on adjacent facilities.	\$50,000
Total Estimated Cost for Project AD2-079		\$750,000

Table 3

Figure 1: Point of Interconnection (Capital Avenue 34.5 kV Substation)
Single-Line Diagram

**AD2-079 Point of Interconnection
Capital Avenue 34.5 kV Substation**

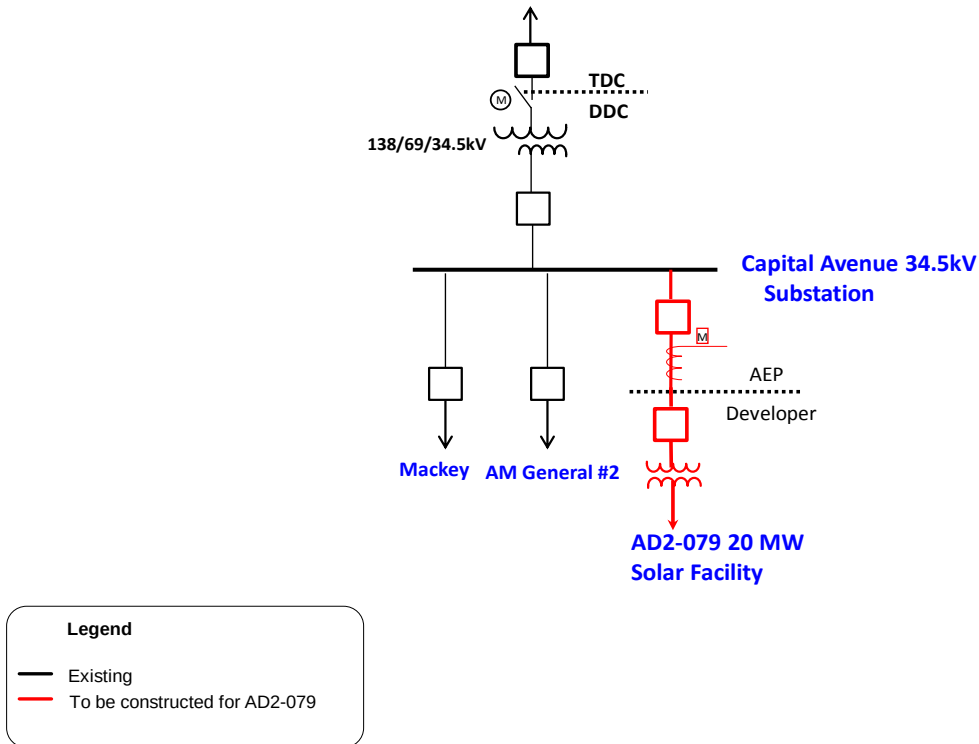


Figure 2: AD2-079 Point of Interconnection (Capital Avenue 34.5kV)

